

## Advancements in Analytical Techniques for Tamsulosin: A Review of Screening Tests and Stability Analysis in Drug Development

**Aniket Rajesing Girase\*<sup>1</sup>, Javesh Kashinath Patil<sup>2</sup>, Devshree Yashwantbhai Patel<sup>3</sup>,  
Harsha Vasudev Chaudhari<sup>4</sup>**

<sup>1,3,4</sup>Student, P.S.G.V.P Mandal's College of Pharmacy, Shahada (Maharashtra), India

<sup>2</sup>Associate Professor, P.S.G.V.P Mandal's College of Pharmacy,  
Shahada (Maharashtra), India

**Corresponding Author**  
**E-mail Id:- aniketgirase@outlook.com**

### ABSTRACT

*Tamsulosin is a medication that men with enlarged prostates or benign prostatic hyperplasia (BPH) can use to treat their symptoms. Creating and certifying screening tests is a crucial phase in the creation of new drugs. This review's primary goals are to give the reader an overview of both new and current drug analysis techniques based on structured raw data and to ascertain whether medications remain stable in the presence of excipients and other stresses that shorten their shelf lives. Because of its great selectivity, specificity, and dependability, this technique can be used for simultaneous drug analysis. The terms accuracy, specificity, precision, limit of quantification (LOQ), limit of detection (LOD), robustness, reliability, and system suitability testing (for a particular drug) are used to define validation parameters. Every test parameter is employed for standard and stability analysis.*

**Keywords:** Prostates, Hyperplasia, Specificity, Quantification, Robustness, Stability Analysis

### INTRODUCTION

Analytical chemistry is the study of methods for determining the chemical composition of samples of substances. A good method provides detailed information about the atomic or molecular species or functional groups present in the sample. Quantitative methods provide numerical data about the balance of one or more of these components.[1] Analytical chemistry research is measured by practical methods and techniques applicable to all fields of science and medicine. He is looking for better ways to determine the chemical composition of natural and synthetic substances. While analytical methods are the specific application of methods to solve analytical problems, chemistry is a discipline that includes a variety of methods in many laboratories.[2] Analytical methods are often developed, refined, and validated

through collaborative research and application. The field of analytical chemistry consists of qualitative and quantitative research. [3,4] Concurrent prediction plays an important role in the pharmaceutical industry because it can be efficient and save time. Various methods such as spectrophotometric methods (UV-VIS, IR, NMR, and mass spectrometry) and chromatographic methods (TLC, HPLC, UHPLC, HPTLC) can be used for many analytical purposes. This method provides a high level of specificity and selectivity and provides a high level of confidence that this method is suitable for the simultaneous analysis of medicinal products. Chromatographic and spectrophotometric methods combined to create new hyphenated methods for simultaneous prediction and impurity analysis. Simultaneous analysis ensures specificity and reliability of identification

of drug components in dosage forms. The main purpose of the evaluation is to determine whether the drug contains the amount of drug specified on the label.[5]

**Analytical chemistry can be divided into two Subfield qualitative analysis and quantitative analysis.**

**Quantitative analysis**

It aims to quantify the quantity of a certain substance contained in a sample. It contains:

**i) Chemical/Classical method:**

These strategies depend on quantifying the appropriate chemical reaction, calculating the amount of reactant required for the overall reaction, or determining the amount of reaction elements obtained.

For example – titrations (acid-base titration, oxidation-reduction titration, non-aqueous titration, complex tissues, acid-base titration, oxidation-reduction titration, non-aqueous titration, complex tissues).

**ii) Instrumental methods:** These methods measure physical or chemical properties of a material, such as its electrical or optical properties, and then correlate these measurements to calculate the concentration of the analyte. Most instrumental techniques fit into one of the three principle areas: Spectroscopy, electrochemical analysis and chromatography.

**Qualitative analysis**

The method of deciding the chemical composition of a test. It incorporates a assortment of explanatory chemistry strategies that give non-numerical information approximately a test. In comparison, quantitative alludes to the quantitative assurance of a test.

**i) Organic qualitative analysis:** Which includes the iodine test, and inorganic qualitative analysis are the two primary divisions of qualitative analysis.

**ii) Inorganic analysis:** Regular inspection of particles in a liquid medium to determine the base and ionic composition of the sample. Natural science research regularly investigates particle types, beneficial agglomerations, and chemical bonds. [6]

**Chromatography**

Chromatography can be a strategy used to separate the components of a mixture by dispersing them sequentially between two steps. One step moves continuously (universal step) over another step (fixed step). [7] The basis of chromatography is the idea that mixtures of molecules applied to surfaces or solids, as well as fluid stationary phases (stable phases), separate from one another while moving with the help of a mobile phase. Molecular properties linked to adsorption (liquid-solid), partition (liquid-solid), affinity, or variations in their molecular weights are among the factors that have an impact on this separation process. These variations lead certain mixture components to travel more slowly through the chromatographic system and remain in the stationary phase for longer than others, while other mixture components enter the mobile phase quickly and exit the system more quickly. Three elements serve as the foundation for the chromatography technique based on this methodology.

- Stationary phase: A "solid" phase, often known as "a layer of a liquid adsorbed on the surface of a solid support," always makes up this phase.
- Mobile phase: "Liquid" or "gaseous component" make up this phase at all times.
- Distinct molecules.[8]

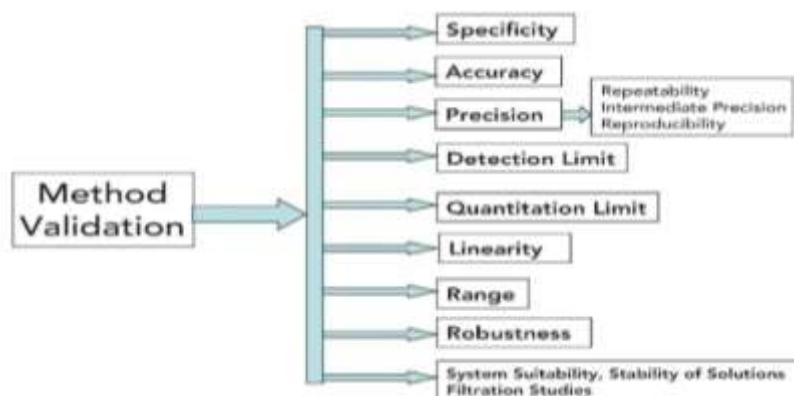
**Validation**

Method validation is an important prerequisite for drug testing. The validation process requires a variety of tests to ensure that the analytical testing

system is suitable for its intended use and can provide good, reliable analytical analysis. As part of the validation process, different methods are tested to determine whether they produce reliable profiles when applied. Verification should include

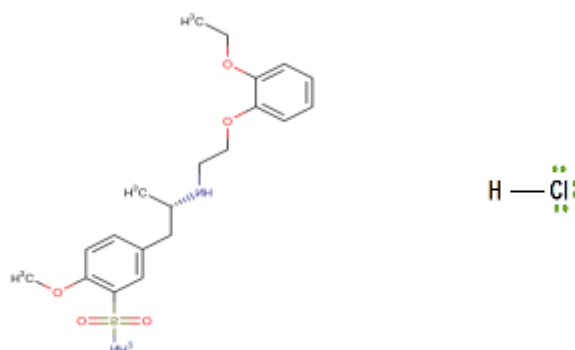
routine configurations, such as additional equipment, to accurately assess the failure process. For this reason, technical validation analyses vary from product to product.[9]

#### *Parameters to be checked for method validation*



#### *Summary Drug Information*

- ✓ **Category** : alpha-1A and alpha-1B adrenergic receptor antagonist
- ✓ **Brand Names** : Flomax, Jalyn
- ✓ **Generic Name** : Tamsulosin
- ✓ **Chemical Name** : 5-(2-[[2-(2-Ethoxyphenoxy)ethyl]amino}propyl)-2-methoxybenzene-1-sulfonamide
- ✓ **Background** : Tamsulosin was first approved by the FDA on April 15, 1997.
- ✓ **Structure** :



- ✓ **Molecular Weight** : Average: 408.512, Monoisotopic: 408.171892706
- ✓ **Chemical Formula** : C<sub>20</sub>H<sub>28</sub>N<sub>2</sub>O<sub>5</sub>S
- ✓ **Melting Point** : 228-230°C
- ✓ **Solubility** : Sparingly soluble in water and methanol, slightly soluble in glacial acetic acid and ethanol, and practically insoluble in ether.[10]

### Mechanism of Action

Tamsulosin is an  $\alpha$ -1A and  $\alpha$ -1D adrenergic receptor blocker. Label, 1 approximately 70% of prostate  $\alpha$ -1 adrenergic receptors are of the  $\alpha$ -1A subtype. Label: By blocking adrenergic receptors, the prostate muscles relax and the outflow of urine is ensured. Blockade of  $\alpha$ -1D adrenergic receptors relaxes the bladder detrusor muscle and prevents symptoms. The specificity of tamsulosin is that it focuses the effect on the target area and minimizes the effect on other areas. [11]

### Evaluation of the Literature

1. **As Per British Pharmacopeia**, Dissolve 3.0 g of sodium hydroxide R in a mixture of 8.7 mL of perchloric acid R and 1900 mL of water for chromatography R; adjust to pH 2.0 with 0.5M sodium hydroxide and dilute to 2000 mL with water for chromatography R; to 1400 mL of this solution add 600 mL of acetonitrile for chromatography R. (As per British pharmacopeia) [12]
2. **As Per Indian Pharmacopeia**, A mixture of 600 ml of acetonitrile and 1400 ml of a solution prepared by dissolving 3.0 g of sodium hydroxide in a mixture of 8.7 ml of perchloric acid and 1900 ml of water, adjusting the pH to 2.0 with 0.5 M sodium hydroxide and diluting to 2000 ml with water. (As per Indian pharmacopeia) [13]
3. **M. Rizk, Zainab M. Mahmoud, Marwa M. Azab et al, (2022)** Tamsulosin hydrochloride and tolterodine tartrate are a binary mixture used in the treatment of benign prostatic hyperplasia. A straightforward, sensitive, stable, and selective thin-layer chromatography method with fluorescence detection was developed for this purpose based on quantitative data from the laboratory. On aluminum paper, silica gel 60 was used for separate thin layer chromatography. The standard was ethyl acetate-n-hexane-diethylamine (9:3:1, v/v). Tamsulosin and

tolterodine had retention factors (RF) of  $0.40 \pm 0.03$  and  $0.85 \pm 0.03$ , respectively. 225 nm was used as the excitation wavelength to detect fluorescence. The range of concentrations for tolterodine and tamsulosin was 100.0 to 900.0 ng/lane and 10.0 to 200.0 ng/lane, respectively. This modification was made in compliance with the International Medication Commission's protocols. Our tamsulosin prescriptions, The purpose of tolterodine and its breakdown products is to be used in high-quality laboratories.[14]

4. **Ahmed SH, Karunakranth D, Babu RS, Khasim SM, Khan MA, et al (2019)**, Prescription drugs contain dutasteride and tamsulosin hydrochloride. Using a mobile phase comprising 40 volumes of a mixture of 20 mM ammonium acetate buffer (pH 3.5): 30 volumes of acetonitrile: 30 volumes of methanol, detected at 223 nm, chromatographic separation was carried out on a C18 column. Tamsulosin hydrochloride ( $r^2 = 0.996$ ) and dutasteride ( $r^2 = 0.998$ ) were found to have a correlation in the range of 24 to 56  $\mu\text{g/ml}$  in the case of drug prices evaluated. The report's suggested procedure is the same as this one. The strategy has been verified. Three retrospective studies were conducted to assess the feasibility of this method at various levels. Additional testing turned up no evidence of the supplement's negative effects. This method was found to be accurate through reproducibility analysis, with an RSD% of less than 2. The method's validity is supported by all statistics, which can be employed for regular prescription history analysis.[15]
5. **Mahmoud M. Sebaiy, Sobhy M. El-Adl et al, (2019)** For the simultaneous determination of tamsulosin and tolterodine in pure foods and pharmaceuticals, an ultrahigh-performance liquid chromatography method has been developed that is fast, sensitive, and accurate. The UPLC technique outperforms the HPLC technique in terms

of cost, weight, speed, and resolution. Using a mobile phase of acetonitrile and 0.025 N, silica core RP C18 (1.7  $\mu$ m, 5 cm  $\times$  2 mm) was separated in an isocratic manner, eluting tamsulosin and tolterodine at 1.54 and 2.43 minutes, respectively. Buffered potassium (50%: 40%). UV detection was at 220 nm, flow rate was 0.5 ml/min, and chromatographic analysis took 5 minutes. The range of tolterodine and tamsulosin is 2–20 mcg/mL. The tablet's data and range are both excellent. The method's linearity, accuracy, precision, specificity, detection, quantification limits, robustness, and reliability were all examined. In compliance with ICH guidelines, the manufacturing process has undergone testing. In the end, a numerical comparison between this model and a reference model revealed that reality and reality are identical.[16]

**6. Vijaykumar et al, (2018),** A first-order differential spectrophotometric method that is straightforward, specific, and accurate was created to analyse tamsulosin hydrochloride and deflazacort simultaneously in a combined preparation. Deflazacort requires a wavelength of 266 nm, while tamsulosin hydrochloride requires a wavelength of 242 nm. For deflazacort, the range was 8 to 40  $\mu$ g/mL, and for tamsulosin hydrochloride, it was 1.6 to 8  $\mu$ g/mL. Deflazacort and tamsulosin hydrochloride have correlation coefficients of 0.9916 and 0.9971, respectively. Tamsulosin hydrochloride and deflazacort had recoveries of 99.31–100.31% and 99.95–100.42%, respectively. Deflazacort and tamsulosin hydrochloride have respective sensitivities of 0.87-1.46% and 1.33-1.87% RSD, and their respective daily sensitivities are 0.86-1.86% and 1.42-1.71% RSD.[17]

**7. Jagdish V.Bharad et al, (2017),** To detect tamsulosin hydrochloride, an easy-to-use, fast, precise, and sensitive ultraviolet

spectrophotometric method was created. With a correlation of 0.9997, this medication has a high absorbance at 224 nm that is linear in the range of 5–25  $\mu$ g/mL. The standards have been confirmed by fact-finding. Three levels (50%, 100%, and 150%) have recovery rates of 50.1%, 99.2%, and 149.2%, correspondingly. This test has been verified for accuracy (repeatability) with a standard deviation of 0.306% and 0.411% (less than 2). Accuracy, precision, linearity, range, accuracy, and repeatability are all acknowledged features of the circuit. Therefore, daily drug quality control can employ the analytical technique for tamsulosin hydrochloride content determination using a UV spectrophotometer.[18]

**8. Eman I. El-Kimary, Essam F. Khamis et al, (2017),** To analyse a combination of tamsulosin hydrochloride (TAM), tolterodine tartrate (TOL), or solifenacin succinate (SOL), and mixed data, a selective thin layer chromatography (HPTLC) method was created. Three compounds are separated using HPTLC, and their spectra are then optically measured at 224 nm as the basis for the suggested procedure. The mobile phase used for the separation was ethyl acetate-methanol-ammonia (6:4:0.05, v/v) on a Merck HPTLC aluminium pan silica 60 F254. The regression lines in the range of 0.1–0.7, 0.4–4, and 1–6  $\mu$ g band-1, respectively, were created using data from linear regression analysis for TAM, TOL, and SOL. The analysis of drug formulations containing two compounds and a mixture ready for testing is done using this validated method. After testing, it was discovered that the model performed well in terms of linearity, accuracy, sensitivity, stability, and precision.[19]

**9. Dr. Venkatesh P, D.Vasavi Devi, Dr. Hepcy Kalarani D, et al (2015),** The new, exact, easy, quick, and accurate method that was suggested was for the development and validation of an



analytical method for the simultaneous estimation of dutasteride and tamsulosin in their combined tablet dosage form using RP HPLC and UV spectrophotometry. Find the absorption maxima of both medications in the UV-visible region in various solvents and buffers, then choose the solvents for the development of the HPLC method. The determination of the dosage forms of dutasteride and tamsulosin hydrochloride tablets was explained using a straightforward and selective HPLC method. With the aid of a mobile phase made up of a mixture of ammonium acetate buffer (pH:3.5), acetonitrile, and methanol (40:30:30 v/v), chromatographic separation was accomplished on a C18 column at a wavelength of 223 nm.[20]

**10. Anamik Shah, et al, (2015),** For the purpose of in-process quality control and the detection of tolterodine and tamsulosin in bulk and capsule formulations, a sensitive, straightforward, and selective HPTLC method has been developed. Aluminum foil-supported silica 60F254 20 × 10 cm thin layer chromatography (TLC) plates with 6 mm bands and 3 µl injection volume were used to separate tolterodine tartrate and tamsulosin hydrochloride. upward flow A chromatography TLC plate was created 30 minutes prior to chamber saturation using a cell that contained methanol, ethyl acetate, and triethylamine in the ratio of 5:5:0.3v/v/v. Tamsulosin hydrochloride and tolterodine tartrate have concentrations of 48–192 ng/bar and 480–1920 ng/bar, respectively. For both medications, the correlation coefficient is 0.999. Tamsulosin hydrochloride and tolterodine tartrate have detection limits of 44.34 ng/lane and 74.85 ng/lane, respectively, while the compounds have limits of 13.26 ng/lane and 22.44 ng/lane. The wavelength at which the optical signal intensity was observed was 220 nm. The process analysis reveals: The system is highly reliable, accurate, and reliable for

its intended use. The solution, specificity, precision, accuracy, detection limit and range, linearity, and reliability of the method are determined.[21]

**11. Reddy et al, (2014),** The goal of the current work was to design and validate a quick and reliable RP-HPLC method that would allow peaks to appear quickly in accordance with ICH Guidelines. Methods: The suggested approach was easy to use, quick, accurate, and precise for both routine quality control analysis and drug quantification in bulk and dosage form. For the simultaneous measurement of Tamsulosin hydrochloride and Dutasteride in bulk drug and combined dosage forms, reversed-phase high-performance liquid chromatography (RP-HPLC) methods were developed and validated. On a Symmetry C18 (4.6 x 150mm, 5µm, Make: XTerra), RP-HPLC separation was accomplished in an isocratic mode. The mobile phase consisted of 80% acetonitrile [HPLC Grade] and 20% phosphate buffer, whose pH was brought to 2.5 using orthophosphoric acid. A flow rate of 0.8 millilitres per minute was observed. 274 nm was chosen as the detection wavelength. There was a 7-minute run.[22]

**12. Makasana Y L , Gotecha N B, et al, (2013),** A method for routinely quantifying finasteride (FINA) and tamsulosin hydrochloride (TAM) in combined dosage form was developed and validated using a precise, straightforward, accurate, and specific ratio derivative UV spectrophotometric technique. Zero order spectra of finasteride and tamsulosin hydrochloride obtained using the first derivative ratio spectrophotometric method were quite both overlapping and parallel. Therefore, it was feasible to estimate both simultaneously using the first derivative ratio spectrophotometry method and these medications. This method can also be used to lessen the matrix effect. Estimation was carried out at peak in the first derivative ratio spectrum. minima of finasteride and

tamsulosin hydrochloride, measured at 253.94 and 235.92 nm, respectively. The calibration graphs showed a linear trend over the concentration range of 1.6-8.0 µg/mL for Tamsulosin Hydrochloride and 10-60 µg/mL for Finasteride. The mean recoveries for Tamsulosin Hydrochloride were 98.05-100.62 % and 98.02-100.79 %, respectively.[23]

**13. Mounika godavarthi, K.sujana, A.pramila rani, et al, (2012),** The reverse phase high pressure liquid chromatographic method for the simultaneous determination of fexofenadine hydrochloride and Montelukast sodium in combined dosage form is described in this research paper as being straightforward, specific, accurate, and precise. A 4.6 x 250 mm column with an internal diameter of 5 µm was used for the separation, and the mobile phase consisted of acetonitrile: TEA (PH 6) (80:20 v/v) flowing at a rate of 1 millilitre per minute. The two medications were separated using RP-HPLC in the absorbance mode at 220 nm. Rt values for FEX and MTKT were  $3.21 \pm 0.01$  and  $6.66 \pm 0.01$ , respectively, indicating a satisfactory resolution of the drugs. In the concentration range of 12-144 µg/ml for FEX and 1-12µg/ml for MTKT, the linear regression analysis data for the calibration plots revealed a good linear relationship with  $r^2=0.9996$  and  $0.9998$  for FEX and MTKT, respectively. The accuracy, specificity, robustness, and precision of the procedure were all confirmed. For FEX and MTKT, the corresponding limits of detection and quantitation were 1.41 and 4.29 µg/ml and 0.02 and 0.06 µg/ml, respectively. FEX and MTKT in bulk and drug formulation can be identified and quantitatively determined using the developed RP-HPLC method that has been proposed.[24]

**14. S. B. Bari et al, (2011),** In accordance with ICH guidelines, a high-performance thin-layer chromatographic method was developed and validated for

the analysis of tamsulosin hydrochloride, both as a bulk drug and in formulations. The method is straightforward, economical, precise, selective, and stable. HPTLC aluminium plates precoated with silica gel 60F-254 were used in the procedure as the stationary phase, and the solvent system contained triethylamine, methanol, and toluene (3.5, 1.2, and 0.2 v/v). The system produced a compact drug spot ( $R_f$  value).of  $0.52 \pm 0.02$ . Tamsulosin was subjected to densitometric analysis at 280 nm using the absorbance mode. The regression that is linear. The calibration plots' analysis data revealed a strong linear relationship ( $r^2 = 0.9982 \pm 0.0012$ ) between the peak area in the range of 400–2400 ng per spot for concentration. Regarding peak are the mean value  $\pm$  SD of the slope and intercept were  $2.6553 \pm 0.0173$  and  $777.7 \pm 74.8$ , respectively. The robustness, precision, and recovery of the method were all verified. The limits for quantitation and detection were 62.10 ng per spot and 20.49 ng, respectively. Tamsulosin underwent thermal degradation, oxidation, and hydrolysis, which show that the medication is heat, oxidation, and hydrolysis susceptible. Statistical analysis demonstrates the method's selectivity, repeatability, and precise in terms of tamsulosin estimation.[25]

## CONCLUSION

The literature study concludes by highlighting several analytical techniques for measuring tamsulosin hydrochloride, also known as tamsulosin HCL, a significant drug for benign prostatic hyperplasia. Techniques described in various pharmacopeias and research papers include spectrophotometric approaches like UV and fluorescence detection, as well as chromatographic techniques like TLC and HPLC. The benefits of these methods are their simplicity, sensitivity, accuracy, and conformity to international standards. Notably, every technique shows

effectiveness when examining tamsulosin hydrochloride alone or in conjunction with other medications such as tolterodine tartrate or dutasteride. The methodologies under discussion offer significant resources for pharmaceutical research and quality control, guaranteeing accurate determination of the tamsulosin hydrochloride concentration in different formulations. Overall, this comprehensive study emphasizes the need of established analytical methodologies in ensuring the quality and efficacy of tamsulosin hydrochloride-based pharmaceuticals.

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